

Henrique Assumpção

Belo Horizonte, Brazil

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Education

MS in Computer Science

Mar 2025 – Jul 2026

Federal University of Minas Gerais, Brazil

GPA: 98%

- **Thesis:** Semidefinite Optimization on Laplacians of Highly Regular Graphs
- **Advisor:** Prof. Gabriel Coutinho

BS in Computer Science

Mar 2020 – Dec 2024

Federal University of Minas Gerais, Brazil

GPA: 94%

- **Thesis:** Algebras, Groups, and Graphs
- **Advisor:** Prof. Gabriel Coutinho
- **Minor:** Mathematics

Professional Experience

Machine Learning Research Scientist

Jun 2025 - Ongoing

Inter Science - Inter&Co., Brazil

Full Time

- Created [CodeEvolve](#), an open-source evolutionary coding agent capable of discovering state-of-the-art algorithmic optimizations across multiple standard benchmarks.
- Lead developer on the *LTM*s project, pioneering the distributed pre-training and optimization of 1B+ parameter Large Language Models on massive multi-modal transactional data for financial forecasting and credit analysis.
- Engineered low-level PyTorch primitives to optimize the distributed training pipeline across multi-node NVIDIA GPU clusters, achieving a 2x increase in token processing throughput.

Machine Learning & Optimization Research Fellow

Dec 2020 - May 2025

DCC-UFMG, Brazil

Fellowship

- Developed *DELATOR*, a scalable GNN framework deployed on production networks exceeding 20M+ nodes and 100M+ edges to capture suspicious transaction patterns missed by rule-based engines.
- Architected a Variational Autoencoder (VAE) in PyTorch for unsupervised anomaly detection on multi-modal industrial time-series telemetry, increasing machine failure prediction precision by 10%.
- Designed a custom recurrent neural network architecture in PyTorch to track longitudinal emotional shifts across large online communities, outperforming established baselines by 20%.
- Derived novel semidefinite optimization bounds for NP-hard combinatorial problems (including MAXCUT and MAX 2-SAT) and leveraged spectral graph methods to model high-dimensional structural representations.

Data Science Instructor

Jun 2022 - Dec 2022

Usiminas / DCC-UFMG, Brazil

Internship

- Designed curriculum and delivered hands-on technical instruction covering production Python, Pandas, and PyTorch paradigms to upskill industrial engineering divisions at a major steel manufacturer.
- Mentored interdisciplinary industrial development teams building internal ML tools, directly driving the automation of legacy data pipelines.

A.I. Research & Development Engineer

Mar 2021 - Aug 2021

Plus Three, USA

Internship

- Engineered end-to-end NLP pipelines using Transformer-based models for complex multi-turn question answering and automated text generation within production client-facing web ecosystems.
- Authored a technical advisory series on algorithmic bias and fairness metrics for the non-profit organization *AlandYou*, translating complex AI safety concepts for cross-functional stakeholders.

Technical Skills

Programming Languages: Python, C++, C, CUDA, Java, JavaScript, Rust

ML Frameworks & Tools: PyTorch, JAX, Triton, TensorFlow, HuggingFace, Scikit-Learn

Infrastructure & Cloud: Docker, AWS, Apache Spark, Linux, Git

Languages: Portuguese (Native), English (C2), Spanish (B2), French (B1)

Papers

Preprints

- **Henrique Assumpção**, Diego Ferreira, Leandro Campos, Fabricio Murai. [CodeEvolve: an open-source evolutionary coding agent for algorithmic optimization](#). *Submitted..*
- **Henrique Assumpção**, Gabriel Coutinho, Chris Godsil. [Total Conformal Rigidity in Graphs](#). *Submitted., 2026.*

Conference Proceedings

- **Henrique S. Assumpção**, Fabrício Souza, Leandro Lacerda Campos, Vinícius T. de Castro Pires, Paulo M. Laurentys de Almeida, Fabricio Murai. [DELATOR: Money Laundering Detection via Multi-Task Learning on Large Transaction Graphs](#). *IEEE International Conference on Big Data (IEEE BigData 2022).*

Journal Articles

- **Henrique Assumpção**, Gabriel Coutinho. [Semidefinite programming bounds on fractional cut-cover and maximum 2-SAT for highly regular graphs](#). *Mathematics of Operations Research*, 2026.
- Bárbara Silveira, **Henrique S. Silva**, Fabricio Murai, Ana Paula C. da Silva. [Predicting user emotional tone in mental disorder online communities](#). *Future Generation Computer Systems*, 2021.

Talks and Presentations

- [CodeEvolve: Evolutionary Agents for Scientific Discovery](#). Invited Talk. [AIDDA 2026](#). Remote.
- [CodeEvolve: an Open Source Evolutionary Coding Agent for Algorithmic Discovery and Optimization](#). Invited Talk. [Inter Science Talks 2026](#). Belo Horizonte, Brazil.
- [Gauge duality for parameters of highly regular graphs](#). Invited Talk. [ILAS 2026](#). Blacksburg, USA.
- [Evolutionary Coding Agents](#). Invited Talk. [Automated Discovery at Scale 2026](#). San Francisco, USA.
- [MAXCUT, Association Schemes and Semidefinite Programs](#). Invited Talk. [CNMAC 2025](#). Rio de Janeiro, Brasil.
- [Primal-dual inequalities for the approximate maximum-cut value of Graphs in Homogeneous Coherent Configurations](#). Poster Presentation. [CNMAC 2024](#). Porto de Galinhas, Brasil.
- [DELATOR: Money Laundering Detection via Multi-Task Learning on Large Transaction Graphs](#). Paper Presentation. [IEEE BigData 2022](#). Osaka, Japan.